

PROMETHEUS FLAME

DRAFT REQUEST FOR SOLUTIONS



Project Title	Prometheus Flame
Project Sponsor	Department of the Air Force
Contracting Activity	Air Force Life Cycle Management Center (AFLCMC)
Questions Deadline	TBD
Response Deadline	TBD
Anticipated Project Budget	TBD
Resultant Award Type	Prizes with follow-on Prototype and Production Other Transaction (OT) Agreements
Authority	10 U.S.C. § 4025, “Prizes for advanced technology achievements.” 10 U.S.C. § 4022, “Authority of the Department of Defense to Carry Out Certain Prototype Projects.”
Request for Feedback	This is a draft solicitation. We welcome your feedback! Submit recommendations NLT 17 April 2026 .

TABLE OF CONTENTS

The Problem.....	2
Our Blueprints for an Open Ecosystem	2
A True Open Competition.....	3
Challenge Phases, Prizes & Fly-Offs	3
The Mission: Your Deliverables (Early Phases)	4
Judging & Evaluation Criteria	4
Step 1: Pre-Fly-off Qualification	4
Step 2: Fly Off	5
Evaluation Factors	5
Mission Scenario	6
How to Enter	6
How do you recommend we improve this challenge?.....	6
Contact Information	7

THE PROBLEM

Open-source solutions like ArduPilot, PX4, CivTAK/ATAK, OpenCV, the Robotic Operating System (ROS), Linux, and Kubernetes have unleashed vibrant ecosystems of innovative commercial capabilities. In defense, progress remains trapped in closed, proprietary "stovepipes" where monolithic software is tightly coupled with bespoke hardware. This prevents the very companies we need—the agile, the innovative, the non-traditional—from contributing to national security.

To kindle an open-source community and ignite innovation at scale, the Mission Systems ASE Group is launching **Prometheus Flame**.

The character of warfare is being reshaped by software. To meet this change, the Department of War (DoW) must revolutionize how it develops and deploys software for Autonomy and Mission Systems applications.

The **Prometheus Flame Challenge** seeks to burn this vendor locked paradigm to the ground and shed light on the potential of open source enthusiasts and non-traditional vendor ecosystems to develop and field standards compliant dual-use Autonomy and Mission Systems product lines. This will create the "flywheel" effect to bootstrap robust commercial and military technologies, lowering cost and increasing capabilities for U.S. and its allies. This initiative directly supports the strategic imperatives to create mass, get humans out of harm's way, accelerate innovations for commercial applications where military dual use exists, and burn down the cost curve by leveraging dual-use Autonomy and Mission Systems capabilities for scalable, affordable effects.

*"Our goal for **Prometheus Flame** is to create an open-source, enthusiast/dual-use-grade learning platform that establishes a new, vibrant community around standards-compliant Autonomy and Mission Systems product lines."*

OUR BLUEPRINTS FOR AN OPEN ECOSYSTEM

To that end, the DoW is publishing two "Government Reference Architecture" (GRA) standards that form the foundation for open, modular, agile, and hardware-agnostic capabilities for next-generation dual-use technologies:

- **Agile Mission Suite GRA (AMS GRA):** Defines a software-defined, modular open architecture for mission system capabilities (e.g., sensors, communication, data fusion, etc.).
- **Autonomy GRA (A-GRA):** Defines standardized behaviors, decision-making, and teaming for autonomous systems.

These standards are currently Controlled Unclassified Information (CUI), however the Government is in the process of downgrading them to Unclassified / Public Release. When publicly released, they will be posted to <https://github.com/modular-af> on or around 1 June 2026. In the interim, Offerors may request the CUI version of the standards by sending an email (see Contact Information section) with documentation that shows they are authorized to handle CUI materials. The Government team will then coordinate the secure delivery of the requested standards. Alternatively, Offerors may download

the publicly available [Open Mission Systems \(OMS\)](#) and [Universal C2 Interface \(UCI\)](#) standards for reference as both AMS GRA and A-GRA's foundation are built upon those standards.

A TRUE OPEN COMPETITION

This is not a typical solicitation. **Prometheus Flame** is a flexible, multi-phase competition designed to identify and fund the best partners to build the foundation of a vibrant new dual-use technology ecosystem.

This is an open competition. **Offerors may enter at any time.** Participation in a prior phase is not required. Whether you are starting with a fresh idea or have an existing prototype, there is an on-ramp for you. At each phase, the goal is to submit as complete a version of the final product as possible for that stage of development.

CHALLENGE PHASES, PRIZES & FLY-OFFS

The **Prometheus Flame** is composed of a continuous challenge effort composed of 3-6 month long phases. Each phase will culminate in a “fly-off” demonstration of each solution in Dayton, Ohio, with the top teams earning a prize.

This RFS will be updated with specific dates once the AMS GRA and A-GRA standards are publicly released. That will start the clock (i.e., T-0) for this challenge.

The first phases of this challenge will focus on building a foundation of open-source software implementation for both AMS GRA and A-GRA. Once that foundation is firmly established, subsequent phases will leverage that to identify best-of-breed Mission Systems and Autonomy skills such as weather management, cooperative search and rescue, etc.

Phase	Date	Per-Team Prize	Num Prizes ¹
1	~T+3 months	TBD	TBD
2	~T+6 months	TBD	TBD
...	...	...	...
N	~T+n*(3 months)	TBD	TBD

Our commitment does not end with the prizes. The **Prometheus Flame** is a valuable mechanism to identify long-term strategic partners. We intend to award Other Transaction Agreements to teams whose solutions are the most compelling and strategically aligned with our vision and long-term needs. This will provide a recurring funding stream to mature the open-source foundation, lead its proliferation, and support the rapidly growing AMS GRA and A-GRA ecosystem. All participants, regardless of the phase they enter, will be evaluated for this opportunity.

¹To qualify for prizes, a solution must prove AMS GRA and/or A-GRA compliance and be delivered with a permissive open-source license (e.g., MIT, Apache, BSD, Unlicense, etc.) for public release. Fewer prizes and OTAs may be awarded if submitted solutions do not meet these criteria.

THE MISSION: YOUR DELIVERABLES (EARLY PHASES)

Our goal for the initial phases of the **Prometheus Flame** is to create an open-source, enthusiast/dual-use-grade learning platform foundation that establishes a vibrant, self-sustaining community—a "flywheel effect"—around standards-compliant Autonomy and Mission Systems product lines. This continuous prize challenge will serve as the primary mechanism to identify and fund the best partners to build the foundation of this new defense ecosystem, enabling hobbyists, academia, non-traditional, and traditional vendors to build dual-use capabilities.

Competitive solutions should include the following public-release content:

- **Open-Source Implementations:** Open-source AMS GRA and A-GRA-compliant implementations that can be simulated and flown on a variety of commercial-off-the-shelf (COTS) hardware solutions. *(Note: This challenge is focused on software development and documentation. It is not intended to mature hardware solutions but rather should leverage existing hardware solutions available on the open market.)*
- **Turn-Key Projects:** End-to-end example AMS GRA and A-GRA-compliant projects (e.g., "hello world," communications, weather management, cooperative autonomous search, etc.).
- **Public-Facing Resource:** A source code repository, public website, getting-started guides, comprehensive documentation, and instructional videos.

Ultimately, the most competitive solutions in the initial phases of this challenge will be those that best empower third parties to readily leverage the provided tools to create their own new, innovative, and standard-compliant products.

JUDGING & EVALUATION CRITERIA

Judging will heavily favor solutions that are not only technically proficient but are also easy to use, well-documented, and extensible. Each phase will culminate in a fly-off event where solutions are evaluated across four critical areas by independent parties. Results will be posted to a public leaderboard.

The evaluation will be completed in two steps: (1) Pre-Fly-Off Qualification & (2) Fly Off

STEP 1: PRE-FLY-OFF QUALIFICATION

This initial qualification step requires the submission of the following materials NLT two months after the Government posts the final solicitation:

- A 5-page / 12-point-font white paper
- AMS GRA and A-GRA compliance report for each Minimum Procurable Units (MPUs) or architectural element developed for the Fly Off.
- A video demonstrating your code running in simulation and in COTS flight-test hardware.

- Hardware and Software Bill of Materials

The Government will evaluate the above materials against the criteria outlined in the “Evaluation Factors” section to select up to 20 submissions for the Fly Off.

STEP 2: FLY OFF

Offerors must provide the following data NLT 24 hours prior to Fly Off:

- All software source code, CI/CD pipelines, and documentation
- AMS GRA and A-GRA compliance report for each Minimum Procurable Units (MPUs) or architectural element developed for the Fly Off
- 5 page / 12-point-font Work Plan for first Project Order of follow-on work (include 3-month base effort, timeline, specific deliverables, and milestone payments)
- Price proposal for the Work Plan (Firm Fixed Price)

The Government will evaluate the above materials against the criteria outlined in the “Evaluation Factors” section to identify Prize and OT awards.

EVALUATION FACTORS

The Government will evaluate Solutions in accordance with the established criteria. Based on the evaluation, the Government will categorize Solutions as “Selectable Plus,” “Selectable,” or “Not Selectable.”

- **Selectable Plus:** Offeror has adequate approach regarding their solution to solve the objectives and is best suited to meet the mission requirement. Highly recommended and prioritized for Fly Off or OT award, subject to available funding.
- **Selectable:** Offeror has adequate approach regarding their solution to solve the objectives.
- **Not Selectable:** Offeror does not have adequate approach regarding their solution to solve the objectives of this Call.

The Government will notify unsuccessful offerors, either via email or letter, to inform them that the proposed effort is not selected at this time.

Category	Description
1. Mission Performance	Simulated and live flight of the software to execute a specific mission scenario. Is it compelling? Does it work reliably? Does it have unique military or dual-use utility?
2. Usability & Onboarding	Evaluation of software, APIs, documentation, and videos to assess how easily a third party can modify or add components. Is the project accessible and well-explained? Are the example projects high quality, varied, and reliable? Do they work “out of the box” and effectively demonstrate key capabilities?

3. Technical Implementation & Standards Compliance	Evaluates the quality, architecture, and AMS GRA/A-GRA compliance of the software itself. Is the code clean, modular, and well documented? Is the simulation environment reliable for pre-flight testing? Is the compliance of the code against the architecture clear and well documented?
4. Risk	Is there a clear path to close on performance in the required timeline? Is the recommended follow-on work compelling? Is the proposed cost for follow-on work appropriate and affordable within the Government budget?

MISSION SCENARIO

TBD. *[We welcome industry’s feedback on what scenario(s) we should leverage for the competition. The scenario should leverage full software-only simulation capability as well as COTS drone/sensor hardware to demonstrate to new users the range of capabilities available in AMS GRA and A-GRA.]*

HOW TO ENTER

Simply send an email (see Contact Information section) with the following information:

- Offeror Name
- Proof of United States or US ally-based headquarters and ownership. This Competition is open to industry and academia.
- Official GitLab account names

The Government will create an Offeror-specific private Git repository where you will post your solutions. Entrance into competition does not guarantee a position at the Fly Off in Dayton, OH.

HOW DO YOU RECOMMEND WE IMPROVE THIS CHALLENGE?

This RFS is still in draft form. NO PROPOSALS are requested at this time. We welcome feedback!

We intend to release the final version of this solicitation on or around 1 June 2026 after the formal public Unclassified release of AMS GRA and A-GRA. That enables multiple rounds of solicitation drafts in the interim for Offerors’ feedback. For this initial draft, **we respectfully request feedback (see Contact Information) no later than 17 April 2026.** We intend to update the draft and provide an additional opportunity for industry to provide feedback before the final solicitation is released. NOTE: Providing responses to this RFI is completely VOLUNTARY. The Government will NOT reimburse any costs associated with providing a response.

Feedback Link: <https://forms.osi.apps.mil/r/TnzeAGVhNh>



CONTACT INFORMATION

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